When Does Context Help Forecasting? An Empirical Study of Energy Systems

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***Abstract*—Energy systems produce operational records alongside power measurements: storage states, charging limits, user requests, and indoor sensor readings. These records can explain why devices with similar recent consumption follow different future trajectories, but their forecasting value depends on both the application and the predictor. We study this question across four public source collections spanning water heaters, electric-vehicle charging, and air conditioning. A shared evaluation task retains query-time information, device identity, and the membership of the population being forecast. Linear, tree, and neural references compare power history, device state, and control information on matched cases. Storage records improve tree and neural water-heater forecasts, while charging declarations improve group forecasts beyond measured telemetry. Indoor measurements yield smaller and model-dependent benefits. A second experiment reassigns operational records within each query population and compares a fixed predictor with retraining under the same reassignment. The substantial recovery on water heaters and one charging site shows why sensitivity of a fitted model and the value of correct record association must be evaluated separately. The study provides source adapters, prepared tasks, and reproducible evaluation code for context-aware energy forecasting.**

***Index Terms*—time-series forecasting, energy systems, operational data, electric vehicles, smart buildings, model evaluation**

I. Introduction

An energy-management system rarely observes only a load curve. A charging station also records energy delivered, charging limits, and sometimes the driver's requested energy and departure time. A water heater reports its operating state and stored energy. A building controller links electricity meters to room temperature and humidity. Collecting and maintaining these records adds complexity, so a practical forecasting question is whether they provide useful information beyond recent power consumption.

The answer matters for both model development and data integration. Two vehicles drawing the same power may have very different remaining energy requirements. Two idle heaters may resume operation at different times because their storage states differ. In these settings, context can resolve ambiguity in the power history. Conversely, a zone sensor shared by several air conditioners may add little to a strong history-based predictor. The presence of an intuitively relevant variable does not by itself establish that a particular model can use it effectively.

The population being forecast is equally important. At a charging site, forecasting demand from vehicles connected now is a different task from forecasting the entire site's future demand, including arrivals. We focus on the former: predict the power trajectories of devices identified at a query time and, where supported, combine their forecasts into the demand of that same group. This exposes a link between device information and aggregate outcomes that disappears when every source is reduced to one load curve.

Energy-forecasting resources already cover rich covariates and several aggregation levels. UrbanEV provides charging-demand data with spatial and external information [1]; the Smarter Mobility Data Challenge studies station, area, and overall occupancy [2]. Device-level forecasting tools also support auxiliary information and appliance behavior [3]. Our study complements these resources by retaining the association between each operational record and the device whose trajectory is predicted, together with the group to which that device belongs.

We organize the evidence around two connected questions. First, when do different kinds of context improve forecasts under fixed model recipes? Second, when a predictor uses linked records, how much of its apparent dependence on those links persists after retraining? The second question requires more than removing inputs at inference. We compare correctly paired records, reassigned records supplied to a fixed predictor, and a predictor retrained on the same reassignment. Power histories and forecast targets are unchanged throughout.

Across water heaters, two workplace charging sites, fast charging, and a Bangkok office building, context yields distinct patterns. Device states help tree and neural models on water heaters; driver declarations provide a larger improvement in charging-group forecasts than the already measured telemetry. Thermal context benefits a linear building-load model slightly but does not improve its stronger tree reference. Retraining recovers much of the reassignment loss in some settings, while another charging site shows little recovery. These differences make the source, information condition, and fitted recipe essential parts of the conclusion.

The contribution is an empirical study and a reusable evaluation task: matched information comparisons across four attributed collections, a record-association experiment connecting fixed-model sensitivity to retraining, and source adapters

that preserve the original device and group semantics. The intended use is to guide choices about contextual inputs and record integration in practical energy forecasting. Code, saved results, and manuscript sources are available online.[1]

II. Related Work

A. Energy data and forecasting tasks

ACN-Data links charging sessions to power, charging controls, and session information [4]. EPFL's fast-charging release records two charging plugs and heterogeneous vehicle sessions [5], while CU-BEMS couples circuit-level power to indoor environmental measurements [6]. BPA's CTA-2045 demonstration provides seasonal water-heater reports and control records [7]. These collections supply different operational views of demand. We retain those differences within one prediction and scoring interface, rather than treating seasons, charging sites, or building floors as independent source collections.

The forecasting task also determines which metadata are usable. Requested charging energy is information about future demand, whereas energy already delivered describes the current state. A realized departure time is an outcome unless it was declared earlier. Our input views distinguish these fields at each query. This permits comparisons among history, state, control, and declarations without changing the scored population.

B. Models and record reliance

We use a regularized linear reference, LightGBM [8], and N-HiTS [9] to examine how the value of context changes across model families. The recipes remain fixed across information conditions. The comparison therefore measures what these concrete predictors obtain from the supplied records, rather than an optimal achievable value of each information source.

Permutation-based analysis evaluates how a fitted predictor relies on its inputs. Fisher *et al.* distinguish reliance of one model from reliance across well-performing models [10]; conditional approaches account for dependencies between retained and perturbed features [11]. Our intervention moves whole auxiliary record blocks, including missingness masks, within the current device group. Pairing evaluation-only reassignment with retraining makes their different consequences visible in device-population forecasting. It is a controlled use of these established ideas, not a new importance estimator.

III. Forecasting from Operational Records

A. Task and information available at a query

At time q, let G_q be the observed device population. For device i, let x_{iq} contain its power history and calendar features, and z_{iq} contain its available operational records and observation masks. A model returns future mean powers

$$\widehat{\boldsymbol{y}}_{iq} = f(x_{iq}, z_{iq}), \qquad \widehat{Y}_{qj} = \sum_{i \in G_q} \widehat{y}_{iqj}. \tag{1}$$

[1]https://github.com/flyingdoog/context-energy-forecasting

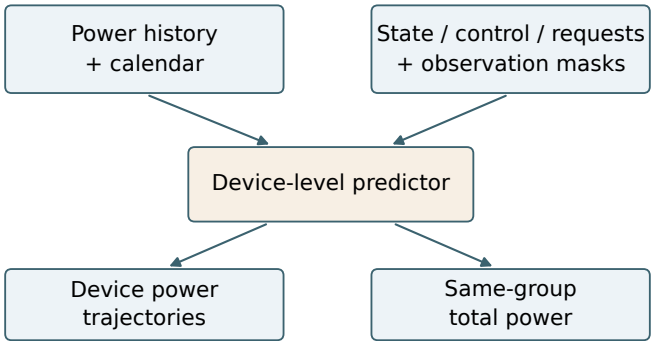


Fig. 1. Context-aware forecasting for the devices observed at a query. Information conditions change the auxiliary inputs, while the power history, device population, and future scoring intervals remain fixed.

The second expression is the group-total forecast for bin j. Its target uses the same membership G_q. Group-mean power divides both sides by $|G_q|$. Device and group errors answer different questions because errors can cancel or reinforce when summed.

We compare three primary information conditions: *history*, *history plus state*, and *all telemetry*, the last additionally including recorded controls. History includes the calendar. At Caltech, a separate comparison adds query-available charging declarations to all telemetry. For CU-BEMS, the state fields are observed temperature and humidity; state and all telemetry coincide because control commands are not supplied. Fig. 1 shows the common task.

All inputs are restricted to information available by the query. Missing records have explicit masks. A confirmed end of a charging session establishes subsequent zero power for that session, but an unobserved tail does not. Charging arrivals after the query are outside G_q. Where a group member has an unsupported future target, the group is excluded from complete-group scoring rather than silently reduced in size.

B. Four source collections

Table I summarizes the prepared evaluation populations. BPA queries occur thirty minutes after a changed-command report, retaining recent power and aligned previous-day/week profiles. Its synchronous groups share the report time and command and contain at least ten label-eligible devices. The measured outcome is reported power, which in this source is partly estimated and calibrated. The groups represent supported observations from the field study, not the entire utility fleet.

ACN uses hourly queries for sessions already present at Caltech or Office. Eligible groups contain at least two distinct charging stations and require complete labels for their query-known cohort. The sites have different information coverage: query-time declarations are observed at Caltech but not in the selected Office records. EPFL contributes 1,771 eligible fast-charging sessions on two plugs. Each query uses the first ten minutes to predict actual delivered power; EPFL is evaluated

TABLE I

SOURCES AND EVALUATION SUPPORT. CASES ARE DEVICE-QUERY PAIRS, EXCEPT EPFL'S ONE QUERY PER SESSION; REPEATED QUERIES ARE NOT INDEPENDENT DEVICES. THE TWO ACN SITES BELONG TO ONE COLLECTION. BOTH LATER PERIODS ARE RETAINED. CU-BEMS COUNTS INCLUDE ALL SUPPORTED FLOORS; TABLE II USES THE MATCHED THERMAL COHORT.

Source / setting	Context beyond power	History	Later period	Cases	Dates
BPA heaters	Storage, flags, commands	24 h + profiles	Spring 2018	64,451	63
			Summer 2018	47,237	69
ACN Caltech	State, pilot, declarations	60 min	October 2019	4,858	31
			January 2020	3,983	30
ACN Office	State, pilot	60 min	October 2019	749	23
			January 2020	796	22
EPFL charging	SoC, energy, capacity, setpoint	First 10 min	March–July 2023	739	100
CU-BEMS ACs	Zone temperature, humidity	60 min	Sep.–Oct. 2019	154,316	61
			Nov.–Dec. 2019	135,839	61

at device level only. Capacity estimates inferred from complete sessions and realized departures are not inputs.

CU-BEMS contains 55 air-conditioning circuits in one Bangkok building, 51 with linked thermal sensor fields. The matched two-period comparison uses 44 circuits on Floors 2–5 and 7 and intersects dates across the compared models. Floor 1 has no thermal fields; Floor 6's published file ends in October. Their available data are retained in the package, but they do not enter this matched thermal comparison. Several circuits can share a zone sensor, so their context is different from a device-specific battery or storage report.

C. Splits, intervals, and error measures

Splits follow time. BPA trains on 46,807 winter cases and validates on the last fourteen observed winter dates, with targets crossing the boundary excluded. ACN trains in April–June 2019 and validates in July. EPFL uses 2022 for training, January–February 2023 for validation, and March–July for evaluation, yielding 946, 86, and 739 sessions. CU-BEMS trains in April–June 2019 and validates in July–August. Table I lists the later periods; no later period is used to tune the reference recipes.

BPA, ACN, and CU-BEMS score cumulative spans of 30, 90, and 150 minutes using fifteen-minute target bins. EPFL scores 5, 15, and 30 minutes using five-minute bins. Source timestamps and the distinction between point reports and interval measurements are retained by the adapters. For a span of J bins, device MSE is the average of $(\widehat{y}_{iqj} - y_{iqj})^2$ over its bins and eligible cases; MAE uses absolute errors. Cases are averaged within each date, and dates then receive equal weight. Span averages weight the three cumulative spans equally.

Power MSE and MAE are reported in kW^2 and kW, respectively. Group scores use the same definitions after aggregation in (1). CU-BEMS additionally weights its common floors equally. We do not pool raw errors across different power scales. Where intervals are quoted below, they are the existing paired calendar-week bootstrap intervals from 1,500 draws, conditional on the fitted predictors.

IV. EXPERIMENTS

A. Reference models and comparison design

Ridge uses an intercept, training-derived standardization, and a unit L2 penalty. LightGBM uses 100 trees, 31 leaves, and squared-error fitting for each target bin. N-HiTS uses historical and current exogenous inputs, known calendar features, and 1,000 squared-error training updates. Neural results average the errors of three independently fitted seeds, not their predictions. Each charging site or building floor is fitted separately.

The recipes are fixed, with no baseline hyperparameter search. Information conditions share the same targets and eligible cases. The neural Caltech declaration comparison keeps its input width fixed by replacing the added fields with zeros in the telemetry-only control; CU-BEMS likewise zeroes thermal slots in the history-only arm. These controls keep a change in available information separate from a change in case selection or network width.

B. Which contextual inputs help?

Table II reports device errors for all three fitted families, every primary information condition, and both later periods where available. The comparison shows that the benefit of context depends jointly on the record type and the predictor.

Stored state helps nonlinear heater forecasts. On BPA, adding state lowers LightGBM MSE from 0.282 to 0.254 in spring and from 0.228 to 0.197 in summer, with lower MAE in both periods. N-HiTS follows the same pattern: 0.322 to 0.297 and 0.274 to 0.237. Adding controls beyond state changes the tree results only slightly. The linear reference does not share these gains. This pattern is consistent with stored state being useful for predicting nonlinear operation, while showing that additional fields alone do not determine the outcome.

Charging telemetry and user intent play different roles. State and control records improve the tree reference at both ACN sites. At Caltech, however, measured telemetry does not substantially lower neural MSE. Declarations provide a clearer improvement because requested energy and intended departure describe demand that has not yet appeared in the power trace. Table III compares these inputs on the same query-known groups. Adding declarations reduces span-average group-mean

TABLE II

DEVICE FORECASTING WITH DIFFERENT INFORMATION CONDITIONS. MSE (KW^2) AND MAE (KW) AVERAGE THE THREE NATIVE SPANS WITH EQUAL DATE WEIGHTS. CU-BEMS ADDITIONALLY WEIGHTS ITS FIVE COMMON FLOORS EQUALLY; ITS STATE AND ALL-TELEMETRY COLUMNS ARE THE SAME THERMAL ARM. NEURAL ERRORS AVERAGE THREE SEEDS. BOLD MARKS THE LOWEST ERROR ACROSS INFORMATION CONDITIONS WITHIN EACH MODEL AND ROW, INCLUDING TIES AT DISPLAYED PRECISION.

Source / period	Model	History		History + state		All telemetry	
		MSE	MAE	MSE	MAE	MSE	MAE
BPA Spring	Ridge	**0.290**	**0.218**	0.314	0.246	0.335	0.273
	LightGBM	0.282	0.192	0.254	0.184	**0.253**	**0.183**
	N-HiTS	0.322	0.208	**0.297**	0.191	0.298	**0.189**
BPA Summer	Ridge	**0.239**	**0.198**	0.256	0.223	0.334	0.290
	LightGBM	0.228	0.176	0.197	**0.166**	**0.196**	0.167
	N-HiTS	0.274	0.194	**0.237**	**0.174**	0.251	0.185
Caltech Autumn	Ridge	**1.183**	**0.484**	1.200	0.498	1.200	0.502
	LightGBM	1.116	0.425	1.103	0.419	**1.093**	**0.415**
	N-HiTS	**1.202**	0.455	1.207	0.440	1.208	**0.437**
Caltech Winter	Ridge	**1.001**	**0.431**	1.043	0.463	1.064	0.473
	LightGBM	1.044	0.385	1.011	0.378	**1.000**	**0.372**
	N-HiTS	**1.075**	0.406	1.080	0.396	1.081	**0.392**
Office Autumn	Ridge	**2.221**	**0.856**	2.588	0.946	2.788	0.997
	LightGBM	2.819	0.893	2.689	**0.886**	**2.665**	0.889
	N-HiTS	3.192	**0.853**	**3.169**	0.859	3.367	0.921
Office Winter	Ridge	**2.623**	**0.937**	2.916	0.992	2.965	1.023
	LightGBM	3.231	1.042	**2.923**	**0.995**	2.942	1.004
	N-HiTS	3.372	**0.933**	**3.316**	0.941	3.525	1.002
EPFL Later	Ridge	583.010	15.910	**529.232**	**14.616**	561.019	15.131
	LightGBM	675.160	17.164	620.133	16.382	**614.475**	**15.959**
	N-HiTS	**740.517**	**17.457**	798.949	18.326	874.857	18.772
CU-BEMS Sep.–Oct.	Ridge	13.784	1.491	**13.630**	**1.460**	**13.630**	**1.460**
	LightGBM	**3.015**	**0.427**	3.317	0.473	3.317	0.473
	N-HiTS	**5.833**	**0.657**	6.420	0.690	6.420	0.690
CU-BEMS Nov.–Dec.	Ridge	12.341	1.429	**12.235**	**1.402**	**12.235**	**1.402**
	LightGBM	**4.111**	**0.488**	4.658	0.558	4.658	0.558
	N-HiTS	**6.871**	**0.698**	7.662	0.763	7.662	0.763

TABLE III

CALTECH GROUP-MEAN FORECASTING WITH QUERY-TIME DEMAND DECLARATIONS. MSE / MAE IN KW^2 / KW; EQUAL-SPAN AVERAGES. THE NEURAL CONTROL HAS THE SAME INPUT WIDTH AS THE DECLARATION ARM.

Model	Period	Telemetry MSE / MAE	+ declarations MSE / MAE
LightGBM	Autumn	0.258 / 0.278	0.209 / 0.251
	Winter	0.256 / 0.290	0.215 / 0.248
N-HiTS	Autumn	0.307 / 0.301	0.229 / 0.261
	Winter	0.325 / 0.318	0.225 / 0.258

MSE by approximately 19% and 16% for LightGBM in autumn and winter, and by 25% and 31% for N-HiTS. MAE also decreases in all four comparisons. Missing declarations remain masked; the comparison does not select only sessions with complete user information.

Fast charging and building sensors give different results. In EPFL, adding state lowers Ridge MSE from 583.010 to 529.232 and LightGBM MSE from 675.160 to 620.133, with corresponding MAE improvements. The neural reference worsens under the same information expansion. In CU-BEMS, thermal inputs yield small gains for Ridge, while the stronger history-based tree has lower errors without them in both later periods. The records here are shared zone conditions, not individual stored-energy states. The distinction is useful operationally: a strong history-only reference provides a meaningful check before expanding the input pipeline.

Across these settings, model choice and information choice cannot be separated. State helps the tree across the heater and charging collections, but the neural gains depend on the particular fields and source. The building example also shows why recording more covariates should be accompanied by a matched information comparison, rather than presumed to produce a better forecast.

C. Correct associations, fixed-model sensitivity, and retraining

The second experiment asks how context is used once it helps prediction. Within a query group, we assign each receiving device an auxiliary record from a deterministic within-group permutation π_q. Power history, calendar, previous-day/week power profiles where available, and targets stay with the receiving device. Auxiliary values and their masks move together. Thus the group retains the same collection of operational records, while individual associations change.

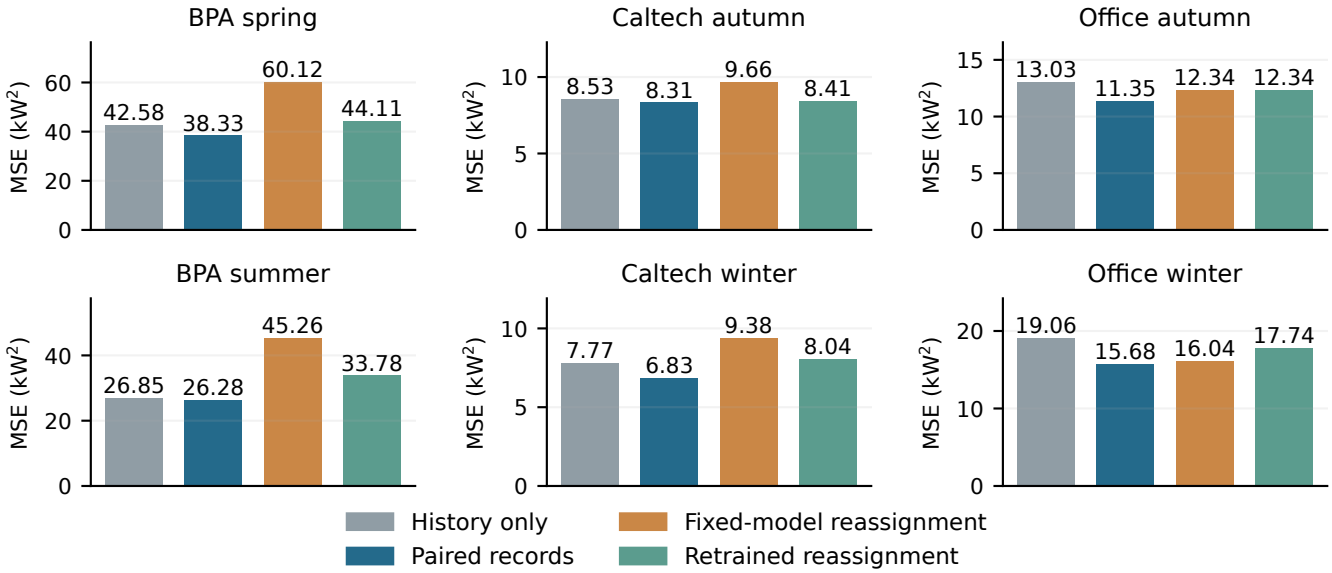


Fig. 2. Group-total MSE for the four record-association arms, averaged over native spans. Both reassigned arms use the same three within-query mappings. Each panel has its own vertical scale. Retraining recovers much of the fixed-model loss in BPA and Caltech, but not in Office.

Four arms use the same fixed LightGBM recipe: history only, paired records, a fitted paired predictor evaluated with reassigned records, and a predictor retrained with reassigned records. Let their risks be R_H, R_P, R_F, and R_R. Two useful differences are

$$D_F = R_F - R_P, \qquad D_R = R_R - R_P. \tag{2}$$

The first measures the fitted predictor's sensitivity; the second measures the remaining gap under retraining with the chosen recipe. Their difference $R_F - R_R$ is the recovered error. Three mapping seeds are used in both reassigned arms, averaging errors across mappings. Constructing a mapping does not read future labels. The experiment uses the ordinary telemetry views, without the extra Caltech declarations.

Fig. 2 retains all six later site-period comparisons. For BPA summer, group-total MSE rises from 26.277 with paired records to 45.256 when only evaluation records are reassigned. Retraining reduces it to 33.780. Relative to the paired error, the fixed-model penalty is 72.23% (95% interval 53.74–94.08%), whereas the remaining retrained penalty is 28.55% (23.86–35.08%). Spring exhibits the same ordering. Correct association is useful for this recipe, but the immediate failure of an unchanged predictor is substantially larger than the gap after adaptation.

Caltech autumn illustrates an even stronger separation. Paired, fixed, and retrained MSE are 8.313, 9.663, and 8.408. The fixed penalty is 16.24% (1.34–32.26%), while the retrained penalty is 1.15% (-5.86–9.69%). In winter, retraining again improves on the fixed arm but leaves a larger gap: 6.827, 9.376, and 8.040, respectively. A fitted model's response to reassignment therefore does not by itself determine how much performance another fit can recover.

Office supplies a different result. In autumn, fixed and retrained MSE are almost identical, 12.343 and 12.341; in winter, retraining increases the error from 16.044 to 17.738. Both later periods still favor the paired reference over history only. Consequently, useful context, fixed-model sensitivity, and successful recovery are three separate empirical properties, not interchangeable descriptions of one effect.

V. Implications and Scope

A. Choosing inputs and maintaining records

The first practical implication is to evaluate information conditions on the same device population. Adding user requests can change which sessions have complete inputs; dropping those with missing requests would also change the forecasting problem. Explicit masks allow the input comparison to retain the original cases. The Caltech experiment then attributes the observed difference to the supplied declarations under its fixed recipes, rather than to a different session subset.

Second, input types should be distinguished by their role. Stored energy, energy already delivered, charging commands, and declared remaining demand are not interchangeable measurements of "state." The declaration results show the value of representing future intent separately from observed telemetry. The building results, in turn, caution against treating a shared environmental measurement as if it were a device-specific state report.

Third, record-association checks should distinguish deployment sensitivity from recovery after updating the predictor. A reassignment test on a fixed model is relevant to an unexpected integration failure. Retraining asks a different operational question: what accuracy does the same learning recipe reach once trained with that representation? BPA and Caltech demonstrate that the answers can differ substantially.

Office demonstrates that retraining is not automatically an effective remedy. Both measurements are needed before converting a model's perturbation response into a claim about the importance of a data linkage.

B. Scope and reusable materials

The evidence covers four collections and three device families, not all energy applications. Seasonal periods and building floors probe different operating conditions within their sources. The charging task concerns devices observed at the query; total future site demand would require an arrival model. EPFL's two-plug source supports session forecasting rather than large-fleet conclusions. Results also condition on the fixed reference recipes and available training data.

The reassignment intervention is synthetic. It preserves within-group record contents but can alter their compatibility with retained histories; it is not conditional resampling given every feature. The measured errors describe predictor behavior under this specified intervention, not the prevalence or causal cost of naturally occurring record errors. These boundaries are particularly important when deciding whether retraining can replace repairing an actual data pipeline.

The repository provides source adapters, a common submission/scoring interface, reference code, and the saved scores used for every table and figure. Preparation produces train/validation/evaluation packets with observation masks and device/group identifiers; the evaluator keeps native intervals and complete-group membership. Source acquisition instructions and attribution accompany the code. Source records and prepared targets are acquired locally under their providers' terms and are not redistributed in the repository.

VI. Conclusion

Operational context can improve energy forecasting, but its benefit depends on what a record describes and how the predictor uses it. Stored state helps nonlinear water-heater forecasts; charging declarations improve group forecasts beyond telemetry; shared indoor measurements yield a different, model-dependent pattern. Record reassignment further shows that the sensitivity of a fitted predictor can greatly exceed the remaining loss after retraining. Matched input comparisons and separate fixed-model and retrained checks provide a practical way to evaluate these effects while preserving the devices, groups, and future intervals of the original forecasting task.

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